

FIRST SOLVE

New here? Do the walkthrough once at cubepath-six-vertex-app/learn. This card is the memory jog.

NOTATION

R U F L D B = turn that face clockwise, looking at it from outside the cube.
 = counter-clockwise. 2 = half turn. Lowercase r l = that face plus the layer behind it, turning together.
 y = spin the whole cube left, white staying down. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns = trigger = a short algorithm your hands run on one turn

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out.
 Can't see it? Build the four white edges on the YELLOW face first, each above its matching center; then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
'RUR'



white sticker FRONT
'L'U'L



white sticker UP run the first one once to knock it out, then look again
'RUR'U'

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.



goes RIGHT
'U' 'R'U' y 'L'U'L



goes LEFT
'U' 'L'U'L y' 'RUR'U'

Edge stick in the wrong slot? Run either one to kick it out, then slide it.

RUR' sexy move RUR' Sune R'FRF' deladeghemmer rap = edge grip

FIRST SOLVE

New here? Do the walkthrough once at [cubepath3x3.vercel.app](#)/learn. This is the memory juggle.

NOTATION

R = turn right face clockwise, looking at it from outside the cube.
 = counter-clockwise, 2 = half turn. Lowercase r = that face plus the layer behind it, turning together.
 y = spin the whole cube, white staying down. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns · trigger = a short algorithm your hands run as one unit
1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it. Can't see it? Build the four white edges on the YELLOW face – first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top; Turn U until it sits above the slot it belongs in, then match a picture below and repeat until it drops in.


white sticker RIGHT
R'UR'


white sticker FRONT
L'ULU


white sticker UR run the first one once to knock it out, then look again
R'UR'

3 MIDDLE EDGES

Top edge with NO yellow; Turn U until its front color matches the center of the edge. The arrow shows which slot it goes to.


goes RIGHT
U' R' L'U'LU


goes LEFT
U' l' L'U'L y' R'UR'

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

RUR sexy move
 RU'R sane
 R'FRF chameleon
 gap = change grip

TWO-LOOK

OLL CROSS - yellow edges



Line bar left-to-right
fRUR'U'f

Hook hook pointing front-right
fRUR'U'f

OLL CORNERS - yellow face



Sune $\frac{1}{2}$ yellow, rest clockwise
RUR'U RU2R'

Anti-Sune $\frac{1}{2}$ yellow, rest counter-clockwise
RU2R' U'RUR'R'



Pi $\frac{1}{2}$ yellow, headlights left
fRUR'U'f fRUR'U'f

OLL CORNERS - continued



Headlights $\frac{3}{2}$ yellow, headlights at the back
R2DR'U2 RD'R'U2R'



2x Headlights $\frac{3}{2}$ yellow, headlights both sides
RUR'U RU'R' URU2R'



Chameleon $\frac{3}{2}$ yellow next to each other
fRUR'U'f **FRB**



Bowie $\frac{3}{2}$ yellow diagonal
fR'URU'r'f **FR**

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked, there is the worst.

Top not all yellow and nothing matches? Run Sun and read again.

No yellow edge at all? Run the first one, then read again.

RUR'U secky move **RUR'U** Sun **R'FRF'** delagehammer rap = change grip

TWO-LOOK

OLL CROSS - yellow edges



Line bar left-to-right
fRUR'U'f'

Hook hook pointing front-right
fRUR'U'f'

OLL CORNERS - yellow face



Sune $\frac{1}{2}$ yellow, rest clockwise
fRUR'U RU2R'

Anti-Sune $\frac{1}{2}$ yellow, rest counter-clockwise
RU2R' U'RUR'



Pi $\frac{1}{2}$ yellow, headlits left
fRUR'U'f' FRUR'U'f'

OLL CORNERS - continued



Headlights $\frac{3}{2}$ yellow, headlits at the back
R2DR'U2 RD'R'U2R'



2x Headlights $\frac{3}{2}$ yellow, headlits both sides
fRUR'U RU'R' URU2R'



Chameleon $\frac{3}{2}$ yellow next to each other
fRUR'U'f' FRF'



Bowtie $\frac{3}{2}$ yellow diagonal
fR'UR'U'f' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked, there is the worst.

Top not all yellow and nothing matches? Run Sune and read again.

No yellow edge at all? Run the first one, then read again.

■ **fRUR'U** sexy move ■ **fRUR'U** Sune ■ **fRFR'** sledgehammer ■ gap = change grip

ONE-LEAF PLI ADJACENT CORNER SWAP	ADJACENT CORNER SWAP
 exactly one face shows headlight	 continued
T 1. headlight; pairs B-left F-left RUR'U' R'F R2UR'U' RUR'F'	Ra 1. headlight; pairs F-left RUR'U' RUR DR'UR'D' R'U2R'
Ja 1. solid; pairs B-right F-left B-front x R2FRF' RU2 R'UR U2	Rb 1. headlight; pairs B-left B-front R2F RURU' F'R U2R'U2R
Jb 1. solid; pairs B-left F-right B-back RUR'F' RUR'U' R'F R2UR'U'	Ga 1. solid; pairs F-right F-back R2UR'UR'U' RUR'2 U'D R'UR'D'
Aa 1. headlight; pairs F-right B-front x L2D2 L'U'L D' L'U'L	Gb 1. headlight; pairs B-back R'UR UD' R2UR'URU' RUR'2D
Ab 1. headlight; pairs B-right B-back x' L2D2 LUL' D'2 LUL	Ge 1. headlight; pairs B-right B-back R2UR'URU' RUR'2 UD' R'UR'D'
F 1. solid R'U'F' RUR'U' R'F R2UR'U'	Gd 1. headlight; pairs B-front B-back RUR' U'D R2UR'UR' U'X'U2R'
One face shows headlight; two corners of that face match. Hold as drawn, run, then turn the top to finish.	Learns in the printed order, three a week, about six weeks. Ja and Jb are the same, then turn the top to finish.
RUR'U' sevey move R2UR'U' Same R'FRF' dedgehammer gap = change grip	

ONE-LOCK PLI

ADJACENT CORNER SWAP

exactly one face shows headlight



- T 1. headlight; pairs B-left F-left
RUR'U' R'F R2UR'U' RUR'F'
- Ja 1. solid; pairs B-right F-left R-front
x R2FRF' RU2 r'Ur U2
- Jb 1. solid; pairs B-left F-right R-back
RUR'F' RUR'U' R'F R2UR'U'
- Aa 1. headlight; pairs R-front x
L2D2 L'U'L D2 L'U'L
- Ab 1. solid; pairs B-right R-back x
L2D2 LUL' D2 LU'L
- F 1. solid
R'U'F' RUR'U' R'F R2UR'U'

One face shows headlights: two corners of that face match. Hold as drawn, run, then turn the top to finish.

RUR'U' easy move **headlights** 2une **R2UR'U'** 3une **R'FRF'** sledgehammer gap = change grip

ADJACENT CORNER SWAP

continued



- Ra 1. headlight; pairs F-right R-back
RUR'U' RU' DR UR'D' R'U2R'
- Rb 1. headlight; pairs R-left R-back
R2F RUR'U' F'R U2RU2R
- Ga 1. headlight; pairs F-right R-back
R2UR'UR'U' RU'R2 U'D R'UR'D'
- Gb 1. headlight; pairs R-back R-front
RUR'U' UD' R2UR'UR'U' RU'R2D
- Ge 1. headlight; pairs R-right R-back
R2UR'UR'U' RU'R2 UD' RU'R'D
- Gd 1. headlight; pairs R-front R-back
RUR'U' UD' R2UR'UR'U' RU'R2D'

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day – one is the mirror of the other.

ANFEX + BIG CUBES AND THE MAP
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BIG CUBES

4x4 and 5x5

Reduce: build the 6 centers → pair the 12 edges → solve it as 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd LAYER ONLY – never widen it.

2 edges

4x4 PLL parity

4x4 OLL parity – 1 edge pair flipped - half the time

4 edge parity – 1 edge pair flipped - half the time

4x4 PLL parity – 1 edge pair flipped - half the time

Fix parity during the last layer, before OLL and PLL – both parity algorithms move corners.

NOTATION



Each letter turns that face clockwise, seen from outside the cube – so from your seat L, D and B look counter-clockwise. • counter-clockwise. 2 = half turn.

U = middle slice: turning the way the face, x y z = the whole cube, turning the way R, D and F turn. Lowercase f = that face plus the layer behind it, turning together.

$Rw \quad U' \quad R' \quad U \quad F' \quad R' \quad F \quad Lw \quad Rw$ slice out, run it, slice back – both cubes
 $2R2 \quad U2 \quad 2R2 \quad Uw2 \quad 2R2 \quad Uw2$ 2 edge pairs swapped - half the time

$Rw \quad U2 \quad x \quad Rw \quad U2 \quad Rw \quad U2 \quad Rw \quad U2 \quad Rw' \quad U2 \quad Rw'$
 $Rw \quad U2 \quad x \quad Rw \quad U2 \quad 3Rw' \quad U2 \quad x \quad Rw \quad U2 \quad Rw' \quad U2 \quad Rw'$

BEGINNER SHORTCUT – use it until you know Sumo
 Twist a yellow corner (four turns at a time, turning ONLY the top face between corners).
yellow faces RIGHT $R'D'R'D \times 2$
yellow faces FRONT $D'R'D'R \times 2$

WORDS

deleteranger = a fixed sequence of turns - trigger = a small algorithm that hands run as one unit - sexy move = the R U R' U' trigger - deledeleanger = the F B F trigger - parity = a case a 3x3 cannot have - edge pair = two edges glued into one, on a 4x4 or 5x5

■ $RUR'U'$ = sexy move
 ■ $RURU$ = Sumo
 ■ $R'F'R'$ = deledeanger
 ■ $rs = \text{csmc grip}$

NNXX – BIG CUBES AND THE MAP DOD A A

BIG CUBES 4x4 and 5x5

Reduce: build the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd Layer ONLY – never widen it.

last 2 edges

4x4 PLL parity

4x4 OLL parity = 1 edge pair flipped - half the time

Rw U2 x Rw U2 Rw U2 Rw U2 U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw U2 Rw' U2 Rw U2 Rw'

5x5 edge parity

Rw U2 x Rw U2 Rw U2 Rw U2 U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw U2 Rw' U2 Rw U2 Rw'

5x5 center parity

Rw U2 x Rw U2 Rw U2 3Rw' U2 U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw U2 Rw'

Fix parity during the last layer, before OLL and PLL – both parity algorithms move corners.

NOTATION

Each letter turns that face clockwise, seen from outside the cube – so from your seat L and D look counter-clockwise = counter-clockwise, z = half turn.

M = middle slice, turning the way L turns. y z = the whole cube, turning the way R, U and F turn. Lower case r f = that face plus the layer behind it, turning together.

BEGINNER SHORTCUT use it until you know Sune

Twist a yellow corner four turns at a time, turning ONLY the top face between corners.

yellow faces RIGHT **R'D'R D** x2

yellow faces FRONT **R'D'R D** x2

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm you hands run as one unit - sexy move = the R U R' U' trigger - slice/turner = the R' F R F' trigger, parity = a case a 3x3 cannot have - edge pair = two edges glued into one, as an 4x4 or 5x5

RUU = sexy move **RUU** = slice **RR'FF** = slice/turner **gap** = change grip

exactly 80 mm: _____

